$\equiv \exists x(\forall y \neg A(x, \text{Kevin Bacon}, y) \wedge \forall y(\neg \exists z A(x,y,z) \vee \neg \exists w A(y, \text{Kevin Bacon}, w)))$ (1st De Morgan's Law)

$\equiv \exists x(\forall y \neg A(x, \text{Kevin Bacon}, y) \wedge \forall y(\forall z \neg A(x,y,z) \vee \forall w \neg A(y, \text{Kevin Bacon}, w)))$ (De Morgan's law of quantifiers)

Some movie actor has not acted with Kevin Bacon in any movie and for any other movie actor, either both of them have not worked together in any movie or the other movie actor has not worked with Kevin Bacon in any movie.

39) Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.

a) $\forall x \forall y (x^2 = y^2 \rightarrow x = y)$

We know, $3 + (-3) = 0$. or, $3 = -(-3)$

$3^2 = (-3)^2 = 9$. $\therefore (3)^2 = (-3)^2$ is True. $3 = (-3)$ is False.

By defn. of conditional, $\boxed{(3)^2 = (-3)^2 \rightarrow 3 = (-3) \text{ is False}}$

By defn. of universal quantifier, $\forall y((3)^2 = y^2 \rightarrow 3 = y)$ is False

By defn. of universal quantifier, $\boxed{\forall x \forall y((x)^2 = y^2 \rightarrow x = y)}$ is False

$x = 3, y = -3$ is a counterexample

b) $\forall x \exists y (y^2 = x)$. We know that $(\pm\sqrt{2})^2 = 2$. Only $\pm\sqrt{2}$ are the two real no.s whose square gives 2. No other no. squared gives 2. $\therefore$ An integer y does not exist s.t. $y^2 = 2$, is True.

By defn. of existential quantifier, $\exists y (y^2 = 2)$ is False. By defn. of universal quantifier, $\forall x \exists y (y^2 = x)$ is False.

$\boxed{x = 2 \text{ is a counterexample}}$.